

William Wang

Software Developer

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Summary

Software Developer specializing in quality engineering and parallel computing platforms, bringing technical experience and a collaborative mindset to solve complex challenges and deliver client-tailored solutions.

Experience

Software Developer – Quality Engineering

May 2024 – Present

IBM Consulting

Toronto, ON

- Designed and implemented quality assurance solutions across multiple layers of the technology stack.
- Automated the upload of test data to SQL databases, leveraging Python libraries to reduce runtime from days to hours.
- Automated the execution of regression tests using IBM's IGNITE tool alongside Java and Selenium.
- Provided technical expertise in Web Content Accessibility Guidelines (WCAG) compliance verification.

Software Engineer Intern – FPGA High Level Design Applications

May 2022 – Sep. 2023

Intel – Programmable Solutions Group

Toronto, ON

- Designed, optimized, and published code demonstrating high-performance parallel algorithms for FPGAs in C++.
- Presented and proposed coding techniques that streamlined user experience and decreased run time by up to 70%.
- Leveraged skills gained from developing libraries and reference designs to accelerate customer adoption and success.
- Led meetings with colleagues across the company to showcase features and gather feedback to drive product focus.

Software Engineer Intern – Test Automation

May 2020 – Aug. 2020

Ciena

Ottawa, ON

- Wrote and maintained Python programs for an elaborate test automation framework with the Control Plane Team.
- Contributed effectively in an Agile environment, actively participating in sprint planning, stand-ups, and retrospectives.
- Expanded the communication of test case results through data visualization components in automated email reports.
- Produced tutorial videos to creatively document features, simplifying complex concepts for a wide range of audiences.

Education

Honours Bachelor of Science – Computer Science Specialist, Mathematics Minor

Sep. 2019 – Apr. 2024

University of Toronto

Toronto, ON

- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Neural Networks and Deep Learning, Artificial Intelligence, Machine Learning, Operating Systems, Parallel Computing, Databases, Web Development.

Projects

Systolic-Array Matrix Multiplication Library | C++, oneAPI, OpenCL, FPGAs

- Designed, optimized, and built a matrix multiplication library that serves as both a reusable library and a reference implementation for linear algebra workloads on FPGAs using High-Level Synthesis.
- Collaborated with compiler engineering teams to integrate advanced compiler features and maximize performance.

AI Search Tool for Legal Rules | Python, Flask, React, MongoDB, OpenAI GPT-3

- Co-engineered an innovative search tool utilizing OpenAI's GPT-3 engine to enable legal teams to retrieve relevant sections of legal documents with greater efficiency and precision.
- Devised a flexible database architecture and backend system that enhanced both development efficiency and scalability.

Skills

Languages: C/C++, Python, Java, SQL, JavaScript, HTML/CSS

Frameworks: OpenCL, oneAPI/SYCL, React, Node.js, Flask, Django, Selenium

Other: Quality Engineering, Customer Experience, FPGA High-Level Synthesis, Parallel Computing Platforms